

SUBHAJIT ROY

 LinkedIn  GitHub  Google Scholar  me.subhajitroy1999@gmail.com  Portfolio

 905 S Dorsey Lane #126, Tempe, AZ, 85281

SUMMARY

Highly motivated PhD candidate specializing in computational biophysics with expertise in DNA nanotechnology, molecular dynamics simulation, and machine learning. Experienced in developing novel computational tools (oxDNA, oxView) and applying deep learning to structural biology problems. Strong track record of publications, cross-disciplinary collaboration, and translating computational insights into experimental validation. Passionate about advancing scientific computing through innovative algorithm development and open-source contributions.

EDUCATION

Physics, PhD

August 2022 - Spring 2027

Petr Šulc's Lab, Arizona State University

CGPA - 3.94/4.00

Computational design and analysis of genetic materials, focusing on DNA origami self-assembly

Physics, Integrated BS-MS

August 2017 - July 2022

UM-DAE-Centre For Excellence In Basic Sciences

CGPA - 7.3/10.0

Master's Thesis: PLAS-5k binding affinity database and retrosynthesis prediction using Deep Learning

Advisor: Prof. Deva Priyakumar, IIIT Hyderabad

RESEARCH EXPERIENCE

PhD Research - Arizona State University

August 2022 - Spring 2027

Advisor: Prof. Petr Šulc

Machine Learning & AI Development:

Skills: PyTorch, TensorFlow, LangChain, CuPy, Transformers, GCNN, Boltzmann Machines, Clustering, Molecular embedding, OpenCV

- Created Scalable Interpolant Transformer with SE(3)-invariant coordinates to generate equilibrated 3D conformations of DNA/RNA from sequence inputs, achieving atomic-level structural accuracy
- Extended transformer architecture to produce relaxed 3D DNA origami structures from line diagrams, enabling biophysics-informed design of complex nanostructures.
- Designed oxView LLM using fine-tuning the Qwen model for end-to-end DNA/RNA structure generation, relaxation, simulation, and analysis through natural language prompts.
- Improved DNA-PAINT super-resolution microscopy to 1nm resolution using DBSCAN clustering algorithms for 3D origami geometry deconvolution
- Integrated oxDNA simulation with ML algorithms to uncover detailed nucleic acid dynamics and folding pathways
- Accelerated AFM image analysis of DNA origami using OpenCV-based automated feature detection.
- Developed LLM-based autograder using LangChain for ASU Physics experimental coursework, reducing grading time by 60% while improving feedback consistency
- Built **fastDNA**: flow-matching generative model (SE(3)-equivariant transformer, Clifford geometric algebra attention) for generating equilibrium 3D DNA/RNA conformations; trained on HPC with PyTorch Lightning and Weights & Biases; model sizes 2M–70M parameters
- Developed **PhysicsGrader / Graviton**: production LLM grading system (LangChain, Next.js, Canvas LMS API) serving 395 students across 10 lab courses; implemented hallucination/factuality checks, RBAC, MFA, and data encryption (ASU compliance)
- Built **NanoCanvas**: cadnano2-inspired DNA origami design tool with full LLM agent REST API (FastAPI, React, Three.js) enabling programmatic structural design via Model Context Protocol (MCP)
- Developed **RAG-backed retrieval pipelines** using LangChain, FAISS vector database, and Hugging Face embeddings for domain-specific scientific document search and context-aware LLM responses

Molecular Simulation & Force Field Development:

Skills: Developer of oxDNA and oxView, AMBER MD, GROMACS, OpenMM, C++, CUDA, Force field development, Cython

- Active core developer of oxDNA and oxView: implemented non-singular mass support, modern GPU compatibility (CUDA), and ARM architecture optimization.
- Co-developed algorithms for inverse design of 3D polycube architectures, determining minimal building block sets and interaction rules for target geometries.

- Built custom CUDA-optimized code to simulate unprecedented system sizes, scaling molecular dynamics simulations to 6 million particles—pushing boundaries of computational feasibility for DNA nanostructure modeling.
- Formulated protein-DNA hybrid system (PROTAC) denaturation pathways using quantum-classical all-atom simulations for targeted protein degradation
- Developed novel coarse-grain force field for irregular origami shapes using rigid body physics and patchy particle models, achieving 1000× speedup over oxDNA2
- Enhanced patchy particle force field for DNA-mediated crystal growth with improved dynamics and phase behavior prediction
- Achieved experimental validation: increased DNA origami crystal growth temperature by 12°C with higher yield based on simulation-guided design
- Designed DNA icosahedron-helix crystal systems with optical properties at visible wavelengths for photonic applications.
- Built oxCloud: unified platform for simulation, analysis, and hosting of DNA origami structure, simplifying the complete pipeline of simulation easier for experimentalists
- Developed Babylon. js-based 3D visualization platform for non-standard coarse-grain molecular models
- Fitted SAX experimental results with simulation using constrained GROMACS simulation.

Computational Infrastructure & Experimental Validation:

Skills: SLURM, Linux Kernel, Docker, Kubernetes, DNA purification, AFM, DNA-PAINT, PCR

- Set up and maintain Linux HPC clusters with x86-64 and ARM64-v8 CPUs and NVIDIA GPUs using SLURM workload manager
- Hands-on experience with experimental techniques: DNA gel electrophoresis, PCR, DNA-PAINT super-resolution microscopy, AFM imaging
- Validated computational predictions through wet-lab experiments, closing the loop between simulation and experimental design

Master’s Thesis - IIIT Hyderabad 2021-2022

Advisor: Prof. Deva Priyakumar

Skills: PyTorch, CNN, AMBER MD, MM-PBSA, MM-GBSA, AutoDock Vina, Molecular docking

- Generated PLAS-5k dataset: 5,000 protein-ligand complexes with binding affinities for ML training (published in Scientific Data)
- Developed CNN model achieving a Pearson correlation coefficient of 0.96 in 10-fold cross-validation
- Automated all-atom MD simulations and extracted binding parameters (MM-PBSA/MM-GBSA), outperforming AutoDock Vina in accuracy and speed
- Published open-access dataset enabling broader community research in drug discovery

Summer Research - IISER Kolkata 2017-2020

Advisor: Prof. Neelanjana Sengupta

Skills: NAMD, Conformational entropy analysis, SASA calculations, Structural persistence analysis

- Investigated allosteric destabilization of ZIKA virus NS1 protein β -ladder domain through disulfide bond reduction
- Performed extensive NAMD simulations, analyzing conformational entropy, enthalpy, SASA, and structural persistence
- Identified biochemical pathways for NS1 destabilization, informing structure-based drug design strategies (published in Biophysical Journal)

TECHNICAL SKILLS

Programming Languages:	Python (PyTorch, TensorFlow, CuPy, Cython, Biopython), C++ (CUDA, OpenMP, MPI), JavaScript (Babylon.js, Three.js, React/Gatsby, Node.js), FORTRAN, MATLAB, Mathematica, Bash, Java, R, C#, Lua
Machine Learning:	PyTorch, TensorFlow, Keras, LangChain, Transformers, GCNN, Boltzmann Machines, Clustering algorithms (DBSCAN), Molecular embeddings, OpenCV, MCP, Hugging Face (fine-tuning, model hub), RAG pipelines, FAISS (vector DB), Weights & Biases, PyTorch Lightning, Flow matching, Diffusion models
Molecular Simulation:	Developer: oxDNA, oxView — Expert: AMBER, NAMD, GROMACS, OpenMM Force field development, Coarse-graining, Enhanced sampling methods
Visualization Tools:	VMD, UCSF Chimera, PyMOL, OxView, Custom Babylon.js/Three.js platforms
HPC & DevOps:	SLURM, Linux kernel, Docker, Kubernetes, Tinycore/EDK2, Git/GitHub
Experimental Methods:	DNA gel electrophoresis, PCR, DNA-PAINT microscopy, AFM imaging, DNA purification, sample preparation
Web Development:	Gatsby (React), Laravel, Lumen, Node.js, React Native, Flutter (iOS/Android)
Hardware/Embedded:	Embedded C (Arduino, STM32), Raspberry Pi, Android (Java), OpenCL

PUBLICATIONS & PREPRINTS

1. **Roy, S.**, Sikand, N., Sample, M., Evans, J., Haggemueller, S., Kikla, A., Matthies, M., and Šulc, P. “OxView, oxCloud and nanobase: an interactive interface for simulation, analysis and sharing of DNA and RNA nanostructures.” *Nucleic Acids Research* (2026, under review).
2. **Roy, S.**, Pradhan, A. K., and Jainarayanan, A. K. ”Molecular Origami in Neuroscience: Noncanonical DNA Structures and Functional Nucleic Acids in Neural Regulation and Disease.” *Frontiers in Neural Circuits* (2026, accepted).
3. Satyabola, D., Feng, S., Chopade, P., Prasad, A., Wang, L., Bamrah, N., Chen, L., **Roy, S.**, Sakai, A., Wang, C., and Sulc, P., 2026. ”DNA-templated Chiral Metamaterial Array as Information Bits.”
4. Diep, T.* , Liu, H.* , Evans, J., Matthies, M., Sample, M., **Roy, S.**, Podbielski, D., Vaidyanathan, S., Williams, D., Kim, K., Kroc, L., Wang, H., Ke, Y., Hihath, J., Yan, H., and Šulc, P. ”Universal 3D Nanoscale Assembly with DNA Polycubes.” *Nature Chemistry* (2026): under review.
5. Zheng, R., Prasad, A., Satyabola, D., Xu, Y., **Roy, S.**, Yan, Y., Sulc, P., and Yan, H. ”DNA-Templated Spatially Controlled Proteolysis Targeting Chimera for Cyclin D1–CDK4/6 Complex Protein Degradation.” *Journal of the American Chemical Society* 147, no. 33 (2025): 29742-29755.
6. Wisna, G.B.M., Sukhareva, D., Zhao, J., Satyabola, D., Matthies, M., **Roy, S.**, Šulc, P., Yan, H., Hariadi, R.F. ”High-speed 3D DNA-PAINT and unsupervised clustering for unlocking 3D DNA origami cryptography.” *Nature Communications*. Nature Communications
7. Korlepara, D.B., Vasavi, C.S., Jeurkar, S., Pal, P., **Roy, S.**, et al. ”PLAS-5k: Dataset of Protein-Ligand Affinities from Molecular Dynamics for Machine Learning Applications.” *Scientific Data* 9, 548 (2022). doi:10.1038/s41597-022-01631-9
8. Roy, P., **Roy, S.**, Sengupta, N. ”Disulfide Reduction Allosterically Destabilizes the β -Ladder Subdomain Assembly within the NS1 Dimer of ZIKV.” *Biophysical Journal* 119, no. 8 (2020): 1525-1537. doi:10.1016/j.bpj.2020.08.036

AWARDS & HONORS

- **NSF Travel Award** (2025) and **ASU Graduate Travel Award** for conference participation
- **DST-INSPIRE Fellowship** - Prestigious national fellowship awarded to top 1% of students in India pursuing PhD in sciences
- **Vijyoshi Science Camp 2018** - Selected participant at IISc Bangalore (organized by KVPY)
- **Teaching & Research Assistantships** - Arizona State University (2022-Present)

PRESENTATIONS & CONFERENCES

- **FNANO 2025** - Presented oxDNA software ecosystem developments and applications
- **Science Leadership Workshop 2020** - Participant
- **Big Data 2020** - Centre for Mathematical Sciences and Applications, Harvard University
- **AWS World Summit Online 2020** - Attended cloud computing and ML sessions
- **Journal of Physical Chemistry Workshop** - IISER Kolkata, June 2018

SERVICE & OUTREACH

- **Member** - Biophysical Society Student Chapter, Arizona
- **Science Outreach** - Presenter at Biodesign Open Door, ASU BioFest 2023 & 2024, Bioscience Network ACP, ASU Homecoming
- **Community Engagement** - Represented ASU Physics Department at Luke Air Force Base Air Day 2024
- **Event Organizer** - Hosted career conversation sessions and BPS student chapter image contest for calendars

ADDITIONAL PROJECTS (DURING BS-MS)

- Skills:** Radio astronomy, Image analysis, Embedded systems, Parallel computing
- Generated and analyzed radio telescope images from GMRT and NASA SkyView across multiple wavelengths (radio, IR, UV, visible)
 - Classified radio galaxies using multi-spectrum data analysis techniques
 - Analyzed real-time radio astronomy data to study Milky Way structure and properties
 - Designed a high-altitude quadcopter using Arduino and STM32 microcontrollers
 - Developed a uniform job distribution algorithm for multi-GPU/CPU systems using OpenCL